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MACHINE LEARNING BASED SPAM EMAIL DETECTION SYSTEM

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Abstract

With the rapid growth of digital communication, email has become one of the most commonly used communication methods. However, the increasing use of email has also led to a rise in spam messages. Spam emails may include advertisements, phishing attempts, and malicious links that can compromise user security and productivity.

This project focuses on developing an automated spam email classification system using machine learning techniques. The Enron Spam Dataset was used for training and evaluating the models. Text preprocessing techniques such as tokenization, stop-word removal, and punctuation removal were applied to prepare the dataset. TF-IDF feature extraction was used to convert textual data into numerical form.

Three machine learning algorithms—Naïve Bayes, Logistic Regression, and Support Vector Machine—were implemented and compared. The models were evaluated using accuracy, precision, recall, and confusion matrix. Experimental results show that the Support Vector Machine model achieved the best performance for spam email detection.

Introduction

Email communication plays an essential role in modern digital communication. Individuals and organizations rely on email for sending messages, documents, and important information.

Despite its usefulness, email systems are frequently targeted by spam messages. Spam emails include unwanted advertisements, fraudulent schemes, and phishing attacks designed to deceive users. These emails not only waste time but can also cause security risks.

Machine learning techniques can automatically detect patterns in email data and classify messages as spam or legitimate. This project aims to build a spam detection system using machine learning algorithms and evaluate their performance on a real-world dataset.

Objectives of the Project

The main objectives of this project are:

• To develop an automated spam email classification system
• To apply natural language processing techniques to email data
• To convert textual data into numerical features using TF-IDF
• To train multiple machine learning models
• To compare the performance of different algorithms
• To identify the most effective model for spam detection

Dataset Description

The dataset used in this project is the **Enron Spam Dataset**, which contains real email messages collected from the Enron Corporation.

Dataset Information

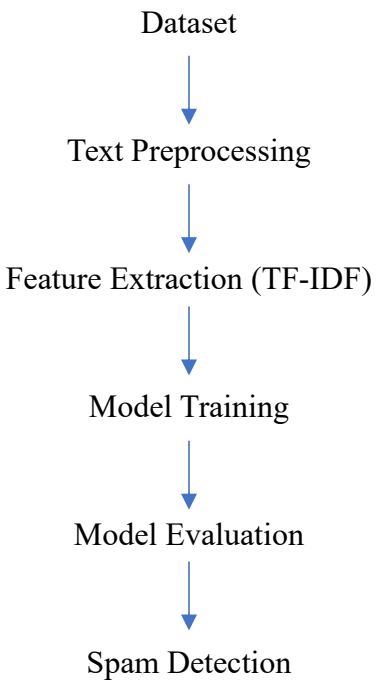
Total emails: **33,716**

Dataset features include:

Column	Description
Subject	Email subject
Message	Email content
Spam/Ham	Email classification label
Date	Timestamp of email

During preprocessing, unnecessary columns such as **Date and index columns** were removed. The **Subject and Message columns were combined** to create a single text field used for model training.

Machine Learning Workflow



This workflow describes the process used in the project. The dataset is first cleaned and preprocessed. The text data is then converted into numerical features using TF-IDF. Machine learning models are trained on the processed data, evaluated using performance metrics, and finally used to classify new emails as spam or legitimate.

Methodology

Data Preprocessing

Before training the models, the email text was cleaned using several preprocessing techniques:

• Converting all text to lowercase
• Removing numbers and punctuation
• Tokenizing the text into words
• Removing stopwords

These steps reduce noise and improve the quality of the input data used for training the machine learning models.

Feature Extraction using TF-IDF

Machine learning algorithms require numerical input. Therefore, the cleaned email text was converted into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency).

TF-IDF assigns higher importance to words that appear frequently in a document but rarely across the dataset. This helps identify keywords that are important for classification.

Configuration used:

TfidfVectorizer(max_features = 5000)

Machine Learning Models

Three machine learning algorithms were implemented:

Naïve Bayes

Naïve Bayes is a probabilistic classification algorithm commonly used for text classification problems. It assumes independence between features and is computationally efficient.

Logistic Regression

Logistic Regression is a supervised classification algorithm used for predicting binary outcomes. It performs well in high-dimensional datasets such as text classification.

Support Vector Machine

Support Vector Machine is a powerful classification algorithm that identifies the optimal boundary separating classes. It is highly effective for text classification tasks.

System Architecture

Email Input

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Text Preprocessing

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TF-IDF Feature Extraction

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Trained Machine Learning Model (SVM)

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Spam / Not Spam Prediction

The system accepts email text as input, processes the text using preprocessing techniques, converts it into TF-IDF vectors, and finally uses the trained machine learning model to classify the email.

Experimental Results

The models were evaluated using the following metrics:

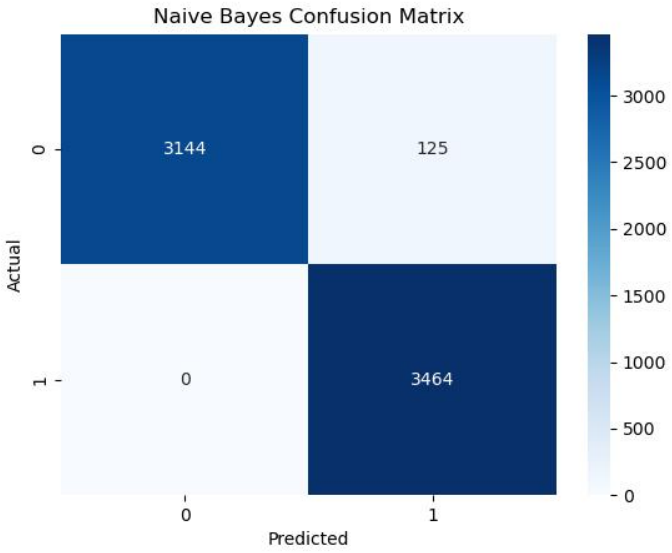
- **Accuracy**
  Accuracy is the percentage of total predictions that the model correctly classified.

- **Precision**
  Precision measures how many of the emails predicted as spam are actually spam.

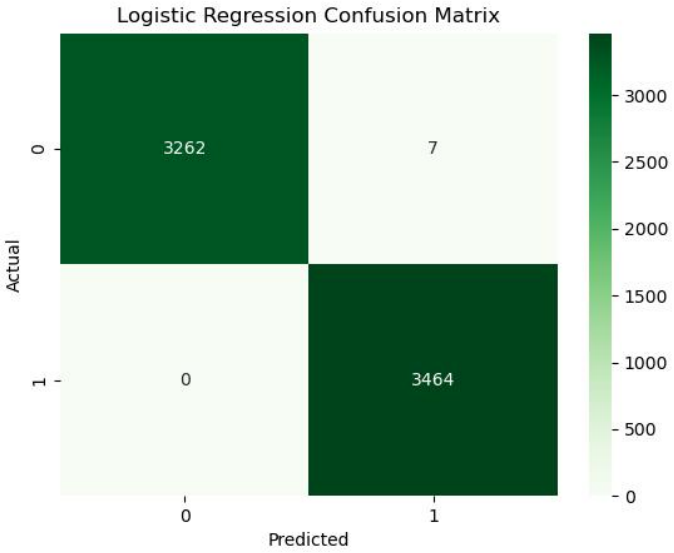
- **Recall**
  Recall measures how many of the actual spam emails were correctly identified by the model.

- **Confusion Matrix**
  A confusion matrix is a table that shows the number of correct and incorrect predictions made by a classification model.

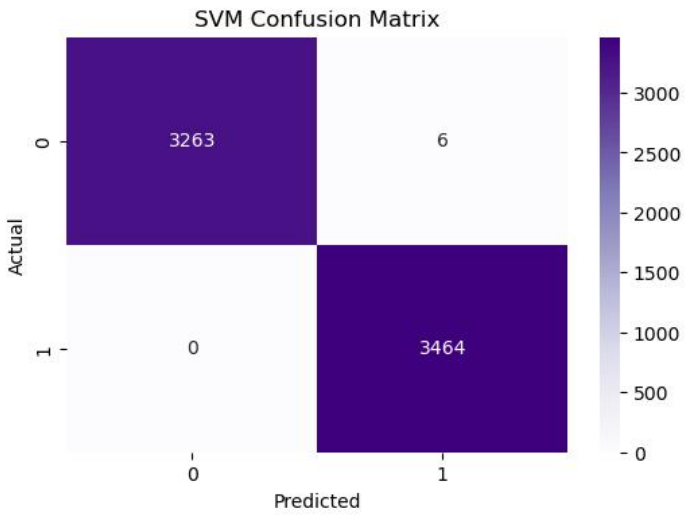
Confusion Matrices



The confusion matrix for the Naïve Bayes model shows the number of correctly classified spam and non-spam emails. It demonstrates that the model performs well but still produces a few misclassifications.

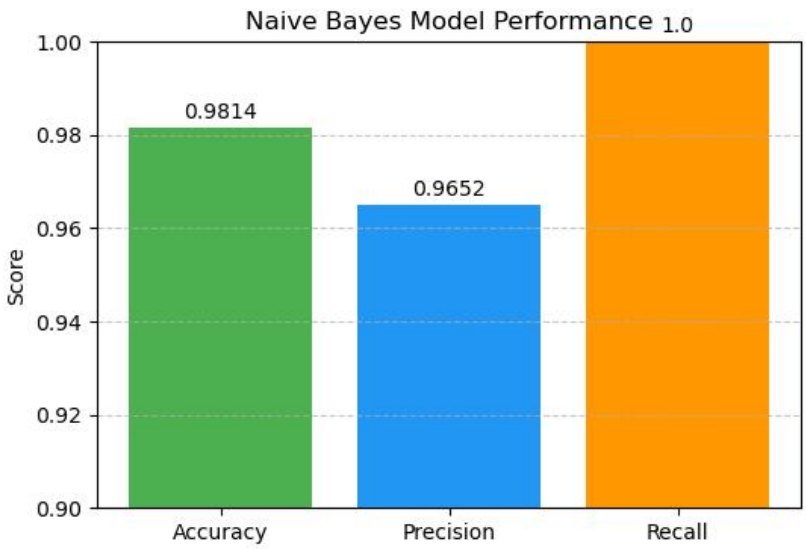


The confusion matrix for Logistic Regression indicates improved classification performance compared to Naïve Bayes. Most emails are correctly identified, indicating a strong classification capability.



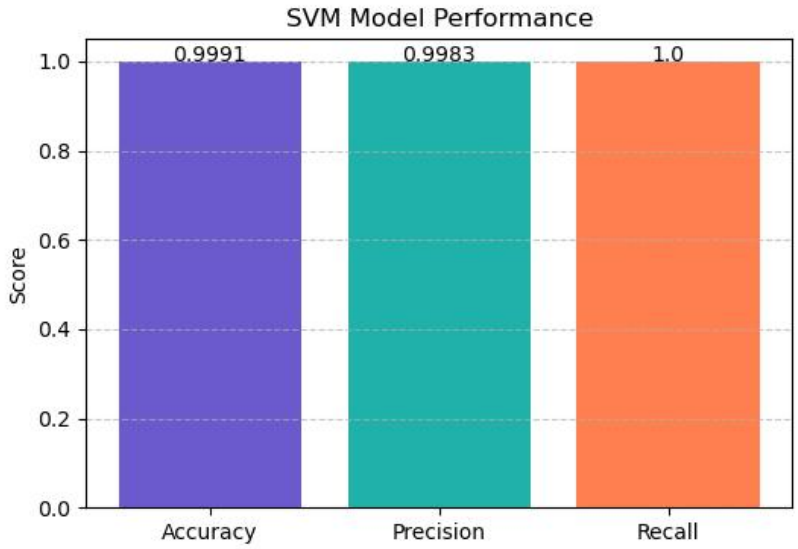
The confusion matrix for the Support Vector Machine model shows the highest accuracy among all models. It correctly classifies almost all spam and legitimate emails.

Model Performance Graphs



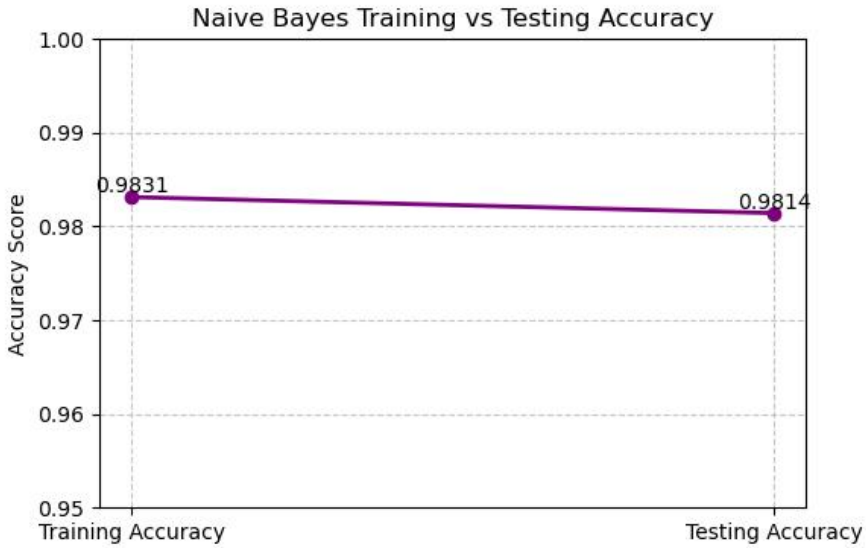
This graph illustrates the performance metrics of the Naïve Bayes model. It shows high accuracy and recall but slightly lower precision compared to other models.

This graph represents the performance metrics for Logistic Regression. The results demonstrate very high accuracy, precision, and recall.

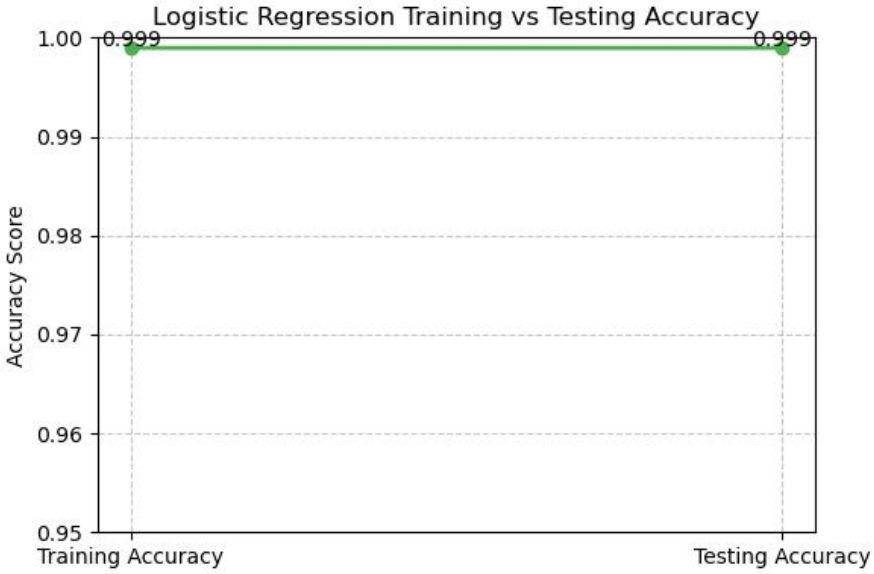


The SVM performance graph shows the highest scores across evaluation metrics, indicating superior performance for spam detection.

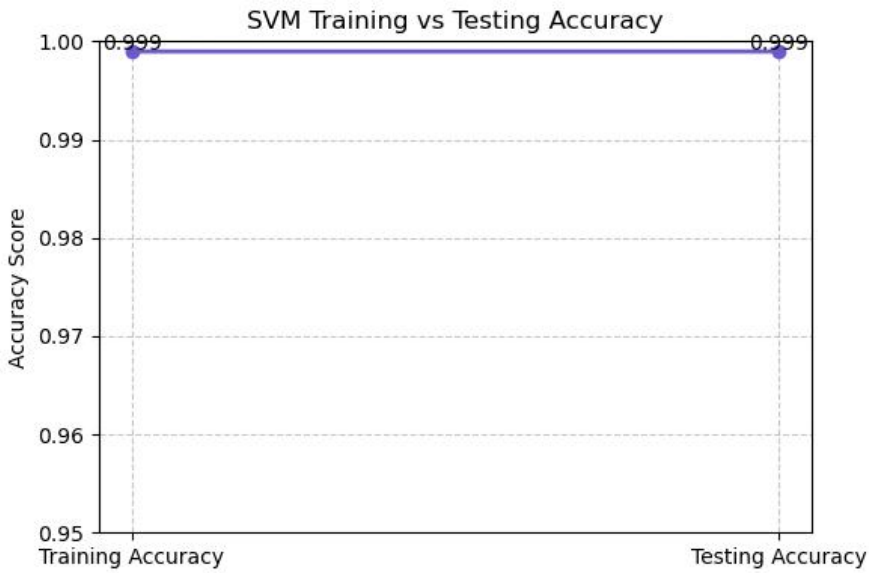
Training vs Testing Accuracy



This graph compares training and testing accuracy for the Naïve Bayes model. The small difference between the two values indicates good generalization.



This graph shows the training and testing accuracy for Logistic Regression. The similar values indicate stable and reliable model performance.



The SVM training vs testing accuracy graph shows almost identical accuracy values, suggesting that the model does not suffer from overfitting.

Spam Detection Demonstration

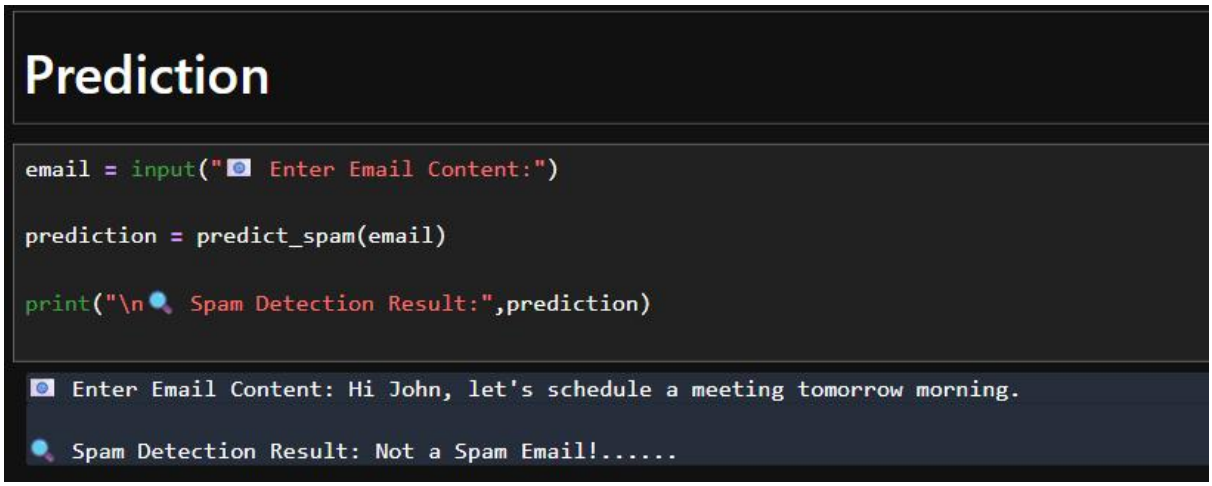
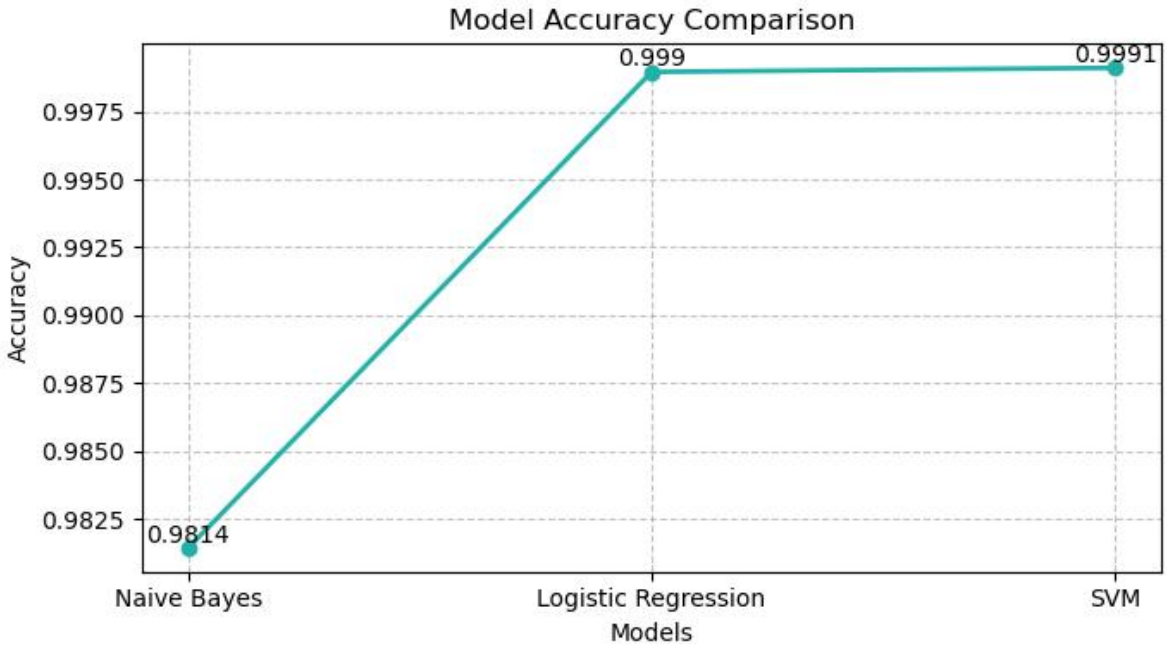


Figure: Spam Email Detection System Output

The system takes an email message as input and processes it using the trained machine learning model. In this example, the email *"Hi John, let's schedule a meeting tomorrow morning."* was correctly classified as Not a Spam Email, demonstrating the model's ability to distinguish legitimate emails from spam.

Model Comparison



This graph compares the accuracy of the three machine learning models used in the project.

Model	Accuracy	Precision	Recall
Naíve Bayes	98.14%	96.52%	100%
Logistic Regression	99.90%	99.79%	100%
SVM	99.91%	99.82%	100%

The Support Vector Machine model achieved the best performance among the evaluated models.

Conclusion

In this project, a spam email classification system was developed using machine learning techniques. The Enron Spam Dataset was used for training and evaluating the models. Text preprocessing and TF-IDF feature extraction were applied to convert email text into numerical features.

Three machine learning algorithms—Naïve Bayes, Logistic Regression, and Support Vector Machine—were implemented and compared. The experimental results showed that the Support Vector Machine model achieved the highest accuracy and provided the most reliable spam detection performance.

The developed system demonstrates the effectiveness of machine learning in identifying spam emails and can be further extended for real-world applications.

Future Work

Possible improvements for the system include:

• Using deep learning techniques such as LSTM or BERT
• Integrating the system with real-time email services
• Expanding the dataset for multilingual email classification
• Developing a web-based spam filtering system

References

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